

Progress Update #2: Metric Analysis & Feature Selection

Gary Wang

Nordling Lab @ Dept. of Mechanical Engineering

National Cheng Kung University

No. 1 University rd., Tainan 70101, Taiwan





Outline

- 1 Metric Orthogonality Analysis
- 2 Feature Selection Results
- 3 Per-Recording Analysis
- 4 Separating Regression from Classification
- 5 Model Comparison Results
- 6 Summary





Outline

- 1 Metric Orthogonality Analysis
- 2 Feature Selection Results
- 3 Per-Recording Analysis
- 4 Separating Regression from Classification
- 5 Model Comparison Results
- 6 Summary





Updated Metric Set

7 metrics analysed:

- MAE ↓
- Spearman ρ ↑
- Weighted κ ↑
- Adjacent Accuracy (AAcc) ↑
- Wasserstein Distance (WD) ↓
- ICC(3,1) ↑
- Accuracy (ACC) ↑

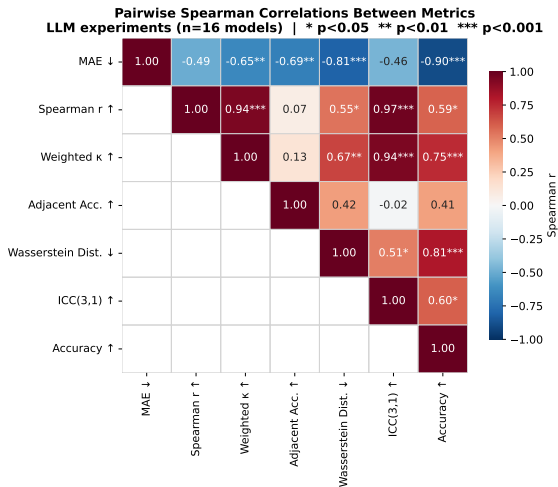
Changes from last meeting:

- Replaced Kendall τ_b with **WD** (more informative for distribution)
- Replaced AUC with **ACC** (task is 5-class, not binary)
- Added **ICC** to LLM summary CSV



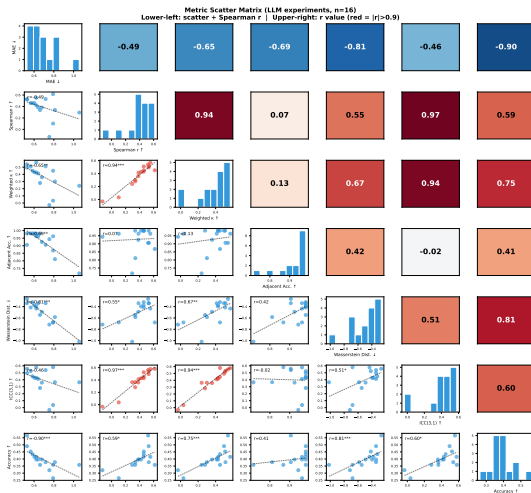


Pairwise Spearman Correlations (LLM, $n=14$ models)



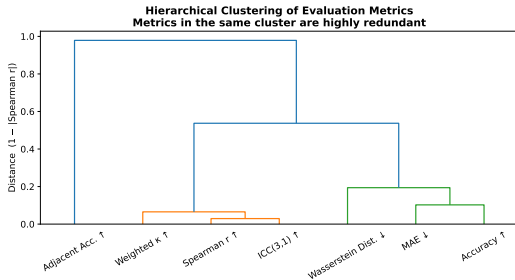


Scatter Matrix – Are Metric Relationships Linear?





Hierarchical Clustering – 3 Groups Identified



3 clusters:

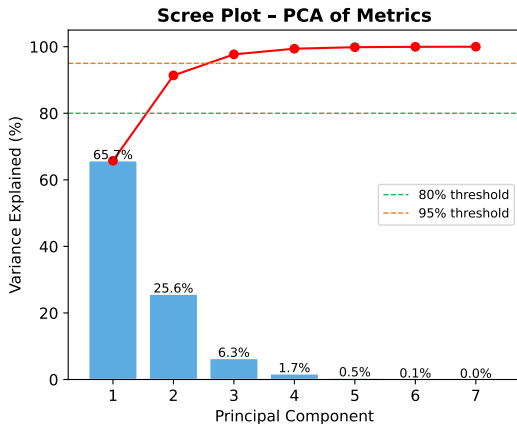
- **Cluster 1:** Spearman, Weighted κ , ICC (*ordinal agreement*)
- **Cluster 2:** MAE, WD, ACC (*error magnitude*)
- **Cluster 3:** AAcc (*most independent*)

⇒ Each cluster picks **one representative**



PCA – PC1 Has Near-Equal Loadings

Principal Component Analysis of Evaluation Metrics
LLM experiments (n=16) — metrics standardised, MAE inverted



PCA Loadings (first 4 PCs)

	PC1	PC2	PC3	PC4
MAE ↓	0.38	0.43	0.13	0.19
Spearman r ↑	0.39	-0.34	-0.38	0.12
Weighted κ ↑	0.44	-0.23	-0.16	-0.01
Adjacent Acc. ↑	0.20	0.63	-0.38	0.42
Wasserstein Dist. ↓	-0.40	-0.30	-0.01	0.84
ICC(3,1) ↑	0.39	-0.39	-0.18	0.09
Accuracy ↑	0.39	-0.07	0.80	0.26



PCA Key Observations

Variance explained:

- PC1: **66.7%** (near-equal loadings 0.34–0.45)
- PC2: 25.8%
- 2 PCs $\geq 80\%$, 3 PCs $\geq 95\%$

Curiosity (for thesis):

PC1 loadings are nearly equal
 $\Rightarrow$ equal-weight composite is a natural aggregate metric
 $\Rightarrow$ but not used by other papers (cannot compare)

Exceptions:

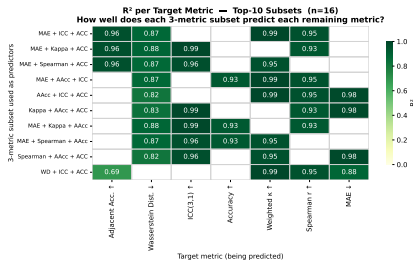
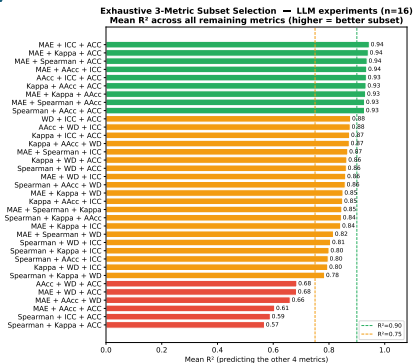
- **AAcc**: loading only 0.21 on PC1
 $\Rightarrow$ most independent metric
- After PC1, AAcc dominates PC2
 $\Rightarrow$ confirms its uniqueness

Conclusion: 3 metrics sufficient
MAE + Weighted κ + AAcc
(one per cluster, most interpretable)





Exhaustive 3-Metric Subset Search



All $C(7,3) = 35$ subsets tried; best subset ($\bar{R}^2=0.957$) linearly predicts the remaining 4 metrics. **AAcc column is always blank** — cannot be predicted by others, confirming its independence.



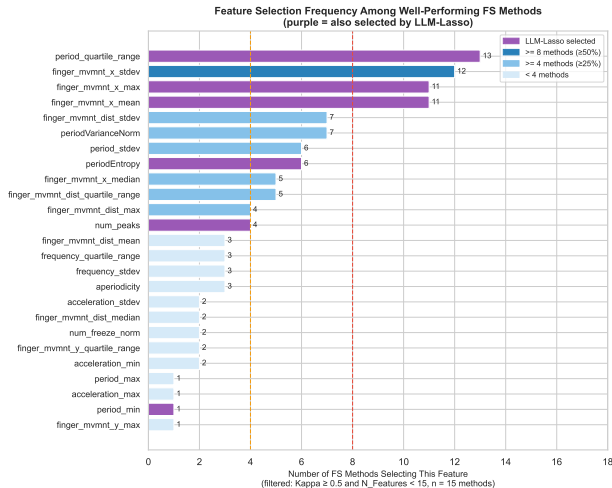
Outline

- 1 Metric Orthogonality Analysis
- 2 Feature Selection Results**
- 3 Per-Recording Analysis
- 4 Separating Regression from Classification
- 5 Model Comparison Results
- 6 Summary





Feature Selection Frequency Across 19 Methods





Most Stable Features

Top features ($\geq 10/19$ methods):

- 1 period_quartile_range
17/19
- 2 finger_mvmnt_x_max
15/19
- 3 finger_mvmnt_x_stdev
13/19
- 4 finger_mvmnt_x_mean
12/19
- 5 periodEntropy
10/19

LLM-Lasso (6 features):

- period_quartile_range[?]
- finger_mvmnt_x_max[?]
- finger_mvmnt_x_mean[?]
- periodEntropy[?]
- period_min
- num_peaks

4/6 in top-5

Standard LASSO (11 features):

- period_quartile_range[?]
- finger_mvmnt_x_max[?]
- periodEntropy[?]
- finger_mvmnt_y_qr
- numFreeze
- frequency_stdev
- num_peaks
- num_freeze_norm
- speed_quartile_range
- speed_min





Outline

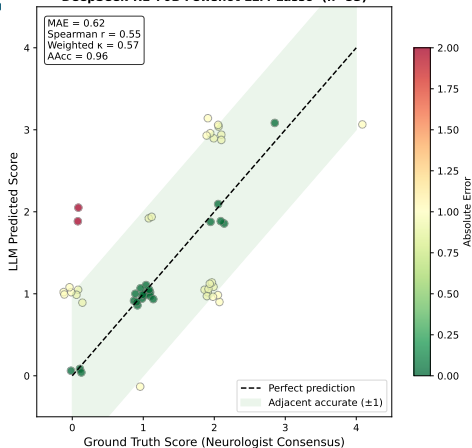
- 1 Metric Orthogonality Analysis
- 2 Feature Selection Results
- 3 Per-Recording Analysis**
- 4 Separating Regression from Classification
- 5 Model Comparison Results
- 6 Summary





Predicted vs Ground Truth (Per Recording)

Per-Recording: Predicted vs Ground Truth
DeepSeek-R1-70B Fewshot LLM-Lasso (n=53)



Model: DeepSeek-R1-70B
Fewshot LLM-Lasso
 $n = 53$ recordings

Observations:

- **96%** of recordings within adjacent accurate band (± 1)
- 2 outliers (red): true score = 0, predicted = 2
- LLM reaches **max prediction = 3**
⇒ underestimates severe cases (score 3–4)
- Regression-to-the-mean tendency for high scores



Outline

- 1 Metric Orthogonality Analysis
- 2 Feature Selection Results
- 3 Per-Recording Analysis
- 4 Separating Regression from Classification**
- 5 Model Comparison Results
- 6 Summary





Borda Count Ranking – All Predictors (LLM-Lasso, 6)

Borda Count Ranking – LLM-Lasso Features (6 features, LOOCV n=53)
Per-metric ranks summed: MAE(M) Kappa(K) AAcc(A) | lower = better

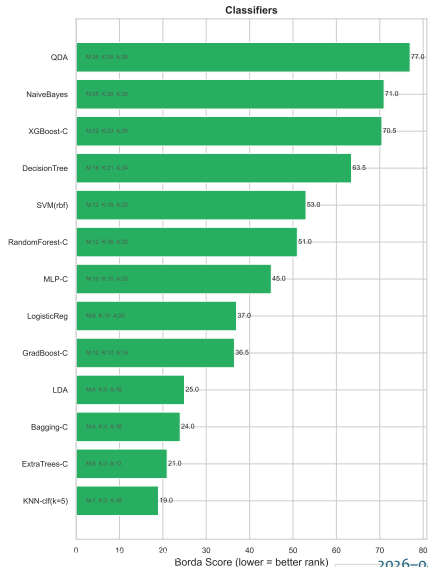
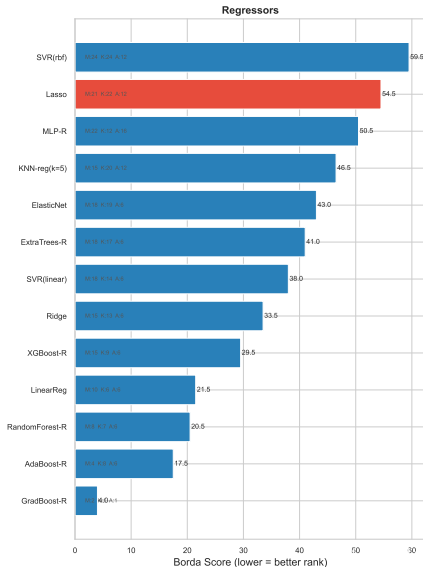


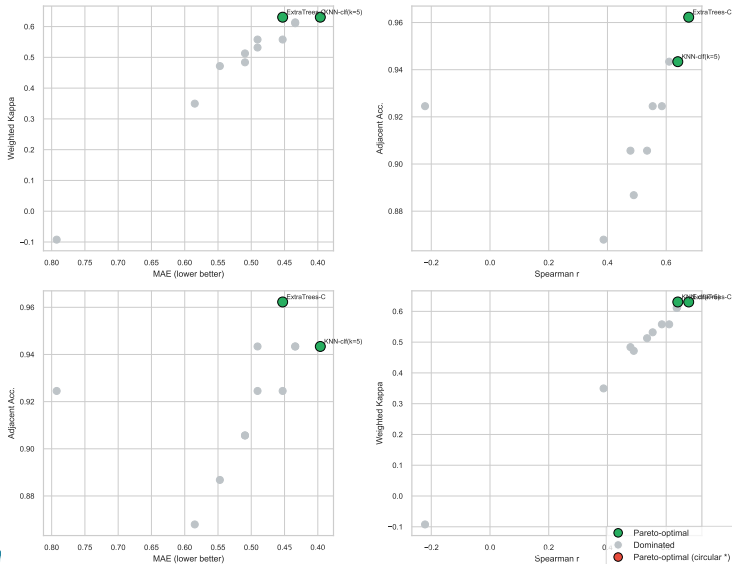
Figure 1 displays four scatter plots comparing Gradient Boosting (GB) and Grid Boosting (GB) models across four metrics: Weighted Kappa, Adjacent Acc., MAE (lower better), and Spearman r. The plots show Pareto-optimal points (blue circles), dominated points (grey circles), and Pareto-optimal points (red circles). The top-left plot shows Weighted Kappa vs MAE. The top-right plot shows Adjacent Acc. vs Spearman r. The bottom-left plot shows Adjacent Acc. vs MAE. The bottom-right plot shows Weighted Kappa vs Spearman r. In all plots, GridBoost-R is the Pareto-optimal point.



Pareto Dominance – Classifiers

Pareto Dominance Analysis – Classifiers (LLM-Lasso, 6 features, LOOCV n=53)

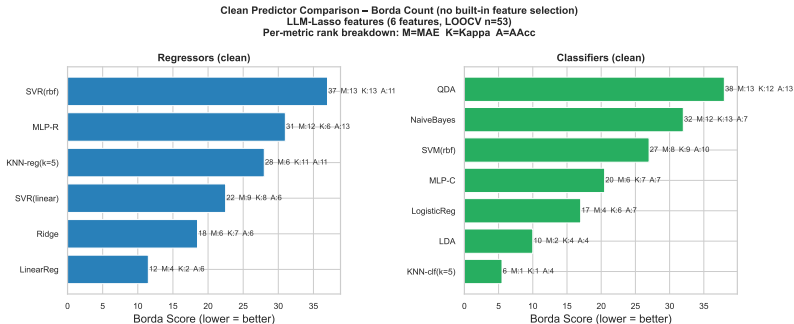
Pareto-optimal: KNN-clf(k=5), ExtraTrees-C





Clean Predictor Comparison (No Built-in Feature Selection)

- Excludes models with internal feature weighting (RF, GradBoost, XGBoost, LASSO...)





Circular Combinations – Flagged with *

- **Definition:** FS method uses the same algorithm as the predictor
 - RF Importance FS + RandomForest-R/C
 - RFECV-RF FS + RandomForest-R/C
 - LLM-Lasso FS + Lasso regressor
- **Problem:** cannot distinguish whether performance is from FS or model
- **Treatment:** included but marked with * and red border in figures





Outline

- 1 Metric Orthogonality Analysis
- 2 Feature Selection Results
- 3 Per-Recording Analysis
- 4 Separating Regression from Classification
- 5 Model Comparison Results**
- 6 Summary





Model Comparison – All Metrics (95% Bootstrap CI)

Model	MAE↓	Spearman↑	Kappa↑	AAcc↑	Stab.↓	Acc Tan↑	Acc Chien↑	Acc Cons.↑
DS-70B FS-LLM	0.62 [0.47, 0.77]	0.55 [0.32, 0.72]	0.57 [0.37, 0.70]	0.96 [0.91, 1.00]	0.380	47.2%	41.5%	41.5%
DS-70B FS-STA	0.57 [0.42, 0.72]	0.46 [0.17, 0.67]	0.51 [0.21, 0.69]	0.98 [0.94, 1.00]	0.252	64.2%	45.3%	48.4%
DS-70B ZS-LLM	0.79 [0.64, 0.94]	0.10 [0.00, 0.23]	0.03 [0.00, 0.08]	0.96 [0.91, 1.00]	0.424	50.9%	33.9%	16.5%
Qw-32B FS-LLM	0.79 [0.60, 0.98]	0.62 [0.41, 0.77]	0.45 [0.28, 0.60]	0.87 [0.77, 0.94]	0.442	35.8%	35.8%	39.3%
Qw-72B FS-LLM	0.53 [0.36, 0.70]	0.52 [0.28, 0.72]	0.54 [0.29, 0.72]	0.91 [0.83, 0.96]	0.340	47.2%	41.5%	56.6%
Qw-72B ZS-LLM	0.75 [0.58, 0.91]	-0.13 [-0.26, -0.10]	-0.03 [-0.11, 0.00]	0.94 [0.87, 1.00]	0.327	54.7%	33.9%	20.0%
Dr. Tan	0.40 [0.26, 0.53]	0.67 [0.48, 0.80]	0.69 [0.50, 0.80]	1.00 [1.00, 1.00]	0.000	—	—	60.4%
Dr. Chien	0.36 [0.23, 0.49]	0.87 [0.79, 0.93]	0.83 [0.73, 0.89]	1.00 [1.00, 1.00]	0.000	—	—	64.2%
Rand. Unif.	1.46 [1.19, 1.75]	0.00 [-0.27, 0.27]	-0.00 [-0.21, 0.21]	0.55 [0.42, 0.68]	—	—	—	—
Rand. Prop.	0.96 [0.75, 1.17]	-0.00 [-0.27, 0.27]	-0.00 [-0.26, 0.26]	0.75 [0.64, 0.85]	—	—	—	—

Best AI model

Baseline





Model Comparison – Normal (0) vs Abnormal (≥ 1)

Model	Bin ACC $\uparrow$	Bin Recall $\uparrow$	Bin Precision $\uparrow$	Bin F1 $\uparrow$
DS-70B FS-LLM	81.1%	97.6%	81.6%	88.9%
DS-70B FS-STA	84.9%	97.6%	85.1%	90.9%
DS-70B ZS-LLM	71.7%	92.7%	76.0%	83.5%
Qw-32B FS-LLM	77.4%	97.6%	78.4%	87.0%
Qw-72B FS-LLM	84.9%	97.6%	85.1%	90.9%
Qw-72B ZS-LLM	75.5%	97.6%	76.9%	86.0%
Dr. Tan	84.9%	100.0%	83.7%	91.1%
Dr. Chien	90.6%	90.2%	97.4%	93.7%
Rand. Unif.	66.4%	80.0%	77.4%	78.6%
Rand. Prop.	64.9%	77.4%	77.3%	77.2%

Best AI model

Baseline



Outline

- 1 Metric Orthogonality Analysis
- 2 Feature Selection Results
- 3 Per-Recording Analysis
- 4 Separating Regression from Classification
- 5 Model Comparison Results
- 6 Summary**





Summary of Updates

1 Metric redundancy confirmed

3 clusters found; recommend **MAE + Weighted κ + AAcc**

2 Feature Selection Results

period_quartile_range most stable (17/19 methods);
LLM-Lasso 4/6 selected features consistent with top-5

3 Per-recording analysis

96% AAcc; underestimation of severe cases is the main limitation

4 Separating Regression from Classification

GradBoost-R Pareto-dominant among regressors;
KNN-clf and ExtraTrees-C Pareto-optimal among classifiers